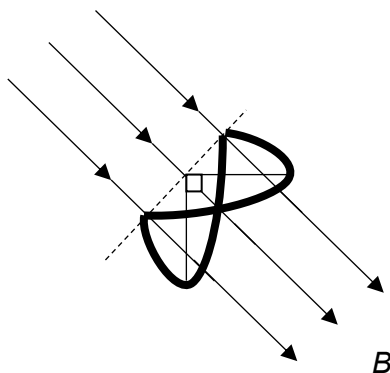


24. A closed circular loop of wire has a radius of 3.7 cm. It is bent along a diameter such that the two halves are perpendicular to each other. A uniform magnetic flux density of $B = 76 \text{ mT}$ is directed perpendicular to the fold diameter and makes equal angles (45°) with the planes of the semicircle.



If the magnetic flux density B is reduced to zero at a uniform rate during a time interval of 4.5 ms, what is the magnitude of the induced e.m.f in the loop?

- A 0.026 V B 0.051 V C 0.073 V D 0.098 V

Ans: B

First we find the effective area that is perpendicular to the field. The magnetic field makes a 45° angle with each half of the semi circle. Thus the effective area is

$$A = 2 \times \frac{1}{2} \pi r^2 \times \cos(45^\circ)$$

Applying Faraday's law for the loop with decreasing magnetic flux:

$$\begin{aligned} \varepsilon &= -\frac{d\phi}{dt} \approx -\frac{\Delta\phi}{\Delta t} \\ &= -\frac{\Delta B}{\Delta t} \times A \quad (N = 1) \\ &= -\frac{0 - 76 \times 10^{-3}}{4.5 \times 10^{-3}} \times \left(2 \times \frac{1}{2} \pi r^2 \times \cos(45^\circ)\right) \\ &= 0.0514 \text{ V} \end{aligned}$$

25 A plane flies through Canada from West to East where the Earth's magnetic field is